

Molecular Analysis of mRNA Vaccine Architecture by Hyperpolarized Solid-State NMR

Research Internship Report

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Abstract

Building on previous work at the CRMN, the aim of the internship was to improve our understanding of the molecular organization of lipid nanoparticles (LNPs) used in messenger RNA (mRNA) vaccine formulations. RNA-based therapeutics rely very often on LNPs to protect the RNA and enable its efficient delivery to target cells, yet their supramolecular architecture remains incompletely understood.

In this work, solid-state Nuclear Magnetic Resonance (NMR) spectroscopy under Magic Angle Spinning (MAS), enhanced with Dynamic Nuclear Polarization (DNP), is employed to investigate the structure of LNPs at the atomic scale. To complement the experimental approach, numerical simulations of proton spin diffusion were developed to evaluate proposed structural models.

Key words : mRNA, Lipid Nanoparticles, Nuclear Magnetic Resonance, Dynamic Nuclear Polarization, Spin Diffusion.

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Contents

Introduction	1
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Part I

Theoretical Background

1 mRNA Medicines and Lipid Nanocarriers	3
1.1 mRNA Based Therapies	3
1.2 Lipid Nanoparticles	4
2 NMR Spectroscopy	6
2.1 Applicability	6
2.2 Basic Principle	7
2.3 Dynamic Nuclear Polarization (DNP)	8
3 Spin Diffusion	10
3.1 Physical Mechanism.	10
3.2 Numerical simulation	11

Part II

Results and Discussion

4 Experimental Results	13
4.1 Material and Methods	13
4.2 Experimental Data Analysis	15
5 Numerical Calculations	18
5.1 Comparison Results.	18
5.2 Beyond Homogeneous LNP Models.	19
Conclusion	22

Appendix

A Chemical Structures	24
B NMR Experimental Setup	25
C Numerical Implementation	27
References	31

Introduction

WHEN it comes to new ways of treating diseases, RNA-based therapies are among the most promising. By regulating gene transcription and protein expression *in vivo*, they offer new treatment strategies for a wide range of diseases, including genetic disorders and cancer.^{[1],[2]} Moreover, this type of therapy benefits from minimizing off-target effects and, hence, side effects during treatment.^[3] An example of RNA-based therapy is messenger RNA (mRNA) vaccines, which has recently become famous in the fight against the global COVID-19 pandemic.^[4] One key factor in the effectiveness of these therapies is the development of efficient delivery systems capable of protecting RNA molecules and transporting them into target cells.^[5]

Among the various delivery systems available, lipid nanoparticles (LNPs) have emerged as leading vectors for RNA delivery. Composed of several self-assembling lipids, these nanostructures are capable of encapsulating and transporting RNA. Even though LNPs are well studied, their supramolecular structure remains poorly understood. In order to rationalize the design and optimization of next-generation RNA-based therapeutics, studies have been conducted to understand the molecular structure and organization of LNPs.^[5]

To investigate the structure of these nanoparticles, a variety of analytical techniques can be employed, such as cryoTEM, SAXS, DLS... Among them, Nuclear Magnetic Resonance (NMR) spectroscopy is very useful to get insight at the atomic scale. NMR spectroscopy uses the magnetic properties of nuclei placed in a magnetic field to study the chemical environment of a given target. In particular, solid-state NMR under Magic Angle Spinning (MAS) has become a powerful tool for studying complex biological systems.^[6] Nevertheless, the widespread application of solid-state NMR is often limited by its intrinsically low sensitivity, resulting in high detection limits and long acquisition times.

In order to overcome this limitation, Dynamic Nuclear Polarization (DNP) has recently been successfully developed.^{[7],[8]} DNP exploits the much larger polarization of unpaired electron spins and transfers it to nearby nuclei of interest, leading to a significant increase in NMR signal intensity. The enhancement of the resulting signal enables the investigation of systems that would otherwise be inaccessible to conventional solid-state NMR.

To better interpret the DNP experiments, numerical simulations of spin diffusion can be useful. These simulations model the transfer of nuclear spin polarization *via* proton spin diffusion. By comparing simulated and experimental data, it becomes possible to evaluate the validity of proposed structural models.^[9]

Following the work of former CRMN PhD student A. Rancz ^[10], the aim of this work is to use solid-state NMR, combining MAS and DNP techniques, to improve the understanding of lipid nanoparticles molecular architecture, in the context of mRNA vaccines. The report is divided into two main parts. The first part consists of a theoretical introduction that sets out the framework of the internship, while the second part presents and analyzes the results obtained.

Part I

Theoretical Background

1 mRNA Medicines and Lipid Nanocarriers

RNA is a fundamental molecule of life. Its role in the living was understood in the mid-20th century as an intermediary “messenger” between DNA (where remains all genetic information) and proteins (which execute most cellular functions).^{[11],[12]} Beyond its central role in gene expression, RNA has rapidly gained a lot of interest as a new therapeutic platform. Indeed, its ability to encode genetic information and direct protein synthesis makes it an excellent candidate for vaccine development for example.^[4]

This chapter briefly introduces the biochemical concepts underlying mRNA vaccine research and focuses on the nanocarriers of interest in this work: lipid nanoparticles.

1.1 mRNA Based Therapies

A bit of Biology...

Definition. Ribonucleic acid (RNA) is a macromolecule composed of ribonucleotides linked by phosphodiester bonds. Each nucleotide contains a ribose sugar, a phosphate group, and one of four nitrogenous bases: adenine (A), guanine (G), cytosine (C), or uracil (U).

RNA is a polyanion due to phosphate groups being charged negatively, that have a significant impact on its interactions with biomolecules and delivery vectors. Unlike DNA, which generally exists as a double-stranded helix, RNA molecules are typically single-stranded and can fold into secondary and tertiary structures through intramolecular hydrogen bonding (base pairing).

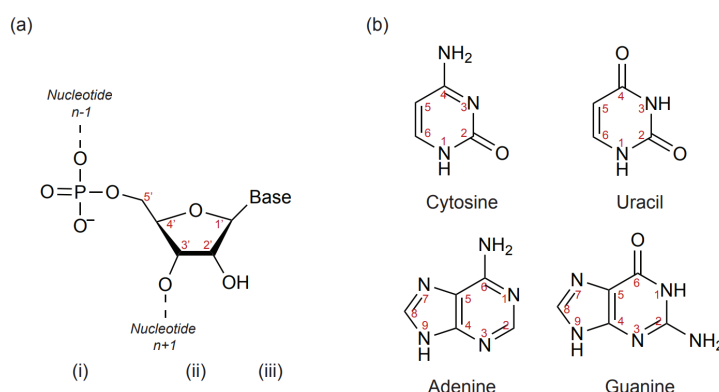


Figure 1.1 : (a) The components of ribonucleotides: (i) the phosphate group, (ii) the ribose sugar and (iii) the base. (b) The four nitrogenous heterocyclic bases (corresponding to (a)/(iii)): the upper ones are the pyrimidine, the lower ones the purine based ones.

There are many types of RNA with different biological functions. Some RNAs are directly involved in protein synthesis, notably mRNA, but also transfer RNA (tRNA), which transports amino acids during translation. Other RNAs play a non-coding regulatory role, such as small interfering RNAs (siRNAs), who are known to be interesting in pharmaceutical research. Here, we will focus on mRNA, which forms the basis of mRNA vaccine.

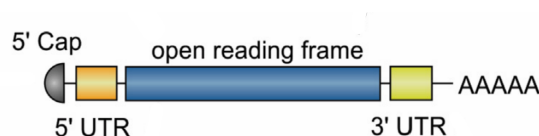


Figure 1.2 : Structural elements of therapeutic mRNA. Each of these elements can be modified to improved the efficiency of the mRNA vaccins.

Figure adapted from [13]

Therapeutic Potential

The therapeutic use of mRNA is based on the delivery of genetic instructions encoding a protein of interest into target cells. Once taken up, the mRNA is translated by the cellular machinery, leading to the production of the desired protein. In the case of mRNA vaccines, cells will produce antigenic proteins that trigger an immune response. [14]

Unlike gene therapies, mRNA molecules only need to enter in the cytosol to be translated: there is no risk of them integrating into the host genome. Furthermore, mRNA-based approaches are perceived to be developed quickly, easily replicated, and readily modified. [13] However, its clinical application has been limited by two major challenges: intrinsic instability and immunogenicity. [4]

- (1) RNA is rapidly degraded by RNases, enzymes that are abundant in biological systems and specifically evolved to cleave RNA molecules.
- (2) Exogenous RNA and/or its delivery system can activate innate immune pathways leading to inflammatory responses that may reduce therapeutic efficacy.

Not to mention the challenge of delivering this large, negatively charged molecule across cellular membranes. This has driven the development of protective delivery systems, among which lipid nanoparticles have emerged as one of the most effective and clinically successful platforms. [5]

1.2 Lipid Nanoparticles

Definition. Lipid nanoparticles (LNPs) are self-assembled lipid-based delivery systems used to protect sensitive therapeutic molecules from degradation, facilitate their entry into body cells, and enable the release onside of it.

Composition of LNPs

LNPs composition may be highly diverse, but most currently clinically relevant systems contain an ionizable lipid as the main component, combined with “helper” lipids.

Ionizable Lipids

Ionizable lipids are pH-dependent lipids that are neutral *in vivo* but become protonated in acidic environments. They typically account for approximately 30–50 mol% of the total lipid content and are essential for the self-assembly of lipid nanoparticles. [15] During formulation, which is usually performed under acidic conditions ($\text{pH} \approx 4$), these lipids become protonated, enabling electrostatic interactions with the negatively charged phosphate backbone of mRNA. Once in the body, their pH-dependent behavior also contributes to endosomal escape, facilitating the release of mRNA into the cytosol.

Helper Lipids

In addition to ionizable lipids, LNPs contain several helper lipids, with some example in Appendix A. These typically include cholesterol, phospholipids, and PEGylated lipids. Using permanently cationic lipids is also possible but they are generally associated with increased systemic toxicity and reduced *in vivo* tolerability. [16] The performance of LNPs depends not only on the individual lipid components, but also on their relative proportions, which influence particle size, stability, vaccine encapsulation, biodistribution, and intracellular delivery. [17]

Formulation of LNPs

Most contemporary LNPs are produced using microfluidic mixing techniques, as illustrated in Figure 1.3. In a typical process, lipids dissolved in ethanol are rapidly mixed with an acidic aqueous buffer containing the mRNA. [18] The sudden change in solvent polarity induces the self-assembly of lipid structures around the vaccine.

Several formulation parameters influence the properties of the resulting nanoparticles, including the lipid composition, the ratio between ionizable lipid and mRNA, the mixing rate, the flow rate ratio between the two phases, and the formulation pH. Finding the right parameters is a research task in its own right. [15]

Once the nanoparticles are formulated, the suspension is purified, for example by dialysis, to remove any residual solvent and replace the formulation's buffer. The resulting LNPs are then generally described as complex supramolecular assemblies with an electron-dense core containing nucleic acids associated with ionizable lipids, surrounded by a shell enriched in helper lipids and PEGylated lipids. [19]

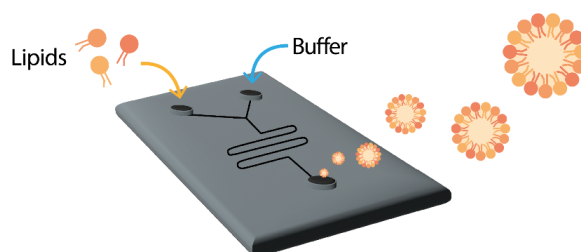


Figure 1.3 : Schematic representation of LNP formation via microfluidic mixing of an ethanolic lipid phase with an aqueous mRNA-containing buffer.

Used with the courtesy of G. Lollo from LAGEPP.

NMR Spectroscopy

NUCLEAR Magnetic Resonance (NMR) spectroscopy is seen as “arguably the most powerful and versatile tool in modern science”.^[20] Exploiting the magnetic properties of matter, NMR can provide local insight on almost any sample of interest. Nowadays, NMR is an indispensable tool not only in fundamental research but also in industrial quality control, chemical analysis, and medical imaging.

This chapter gives a fairly broad overview of NMR basics, before taking a closer look at the hyperpolarization technique used during this internship: Dynamic Nuclear Polarisation (DNP).

2.1 Applicability

Unlike diffraction methods, which require long-range ordered (non amorphous) structures, NMR does not have the same limitation and is applicable to both ordered and disordered systems, while also being non-destructive.^[21]

The NMR Hamiltonian contains isotropic and anisotropic terms, meaning dependent (or not) on the spatial orientation of the molecules in the sample. In solution-state NMR, the rapid molecular motion averages the anisotropic terms to zero, leaving only isotropic contributions, thus giving high high-resolution spectra. In solids, molecular motion is typically insufficient to average these interactions, leading to broad and most of the time useless NMR spectra.^[6]

Fortunately, the first-order anisotropic interactions all share the same angular dependence, proportional to $3 \cos^2 \theta_m - 1$, with θ_m the so-called “Magic Angle”. The sample is spun around an axis oriented at $\theta_m \approx 54.7^\circ$ with respect to the external magnetic field, as illustrated in [Figure 2.1](#): it’s the Magic Angle Spinning (MAS) method. When the spinning frequency is sufficiently high compared to the anisotropic interaction strengths, these first-order anisotropic interactions are averaged out, producing significantly narrower spectral lines and useful spectra.^[22]

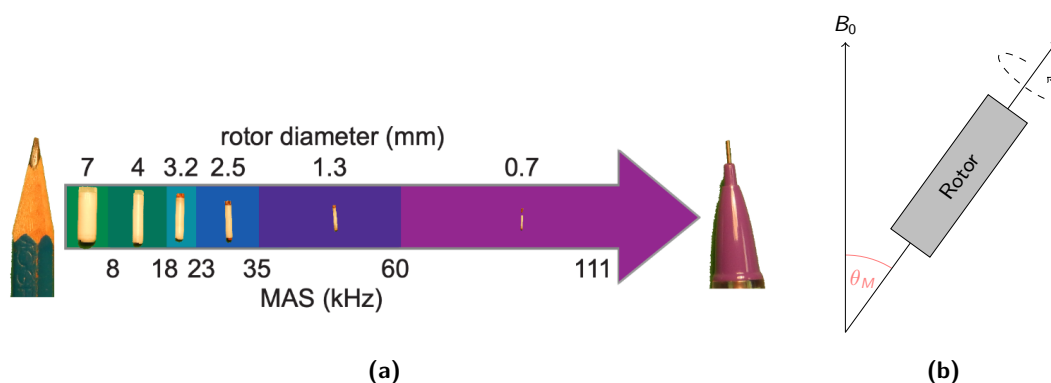


Figure 2.1 : (a) Photos of various available rotor sizes, with the maximum accessible rotations speeds. (b) MAS diagram : rotor spinning at the magic angle ($\theta_M \approx 54.7^\circ$) with respect to B_0 .

(a) Used with the courtesy of G. Pintacuda from CRMN.

2.2 Basic Principle

Definition. Nuclear magnetic resonance (NMR) spectroscopy is a technique based on the oscillatory response of certain nuclei when placed in an external magnetic field B_0 and excited by radiofrequency radiation.

Historically, the first observation of nuclear magnetic resonance phenomena was made by Rabi and co-workers in 1938 using molecular beam experiments.^[23] They showed that an energy splitting occurs:

$$\Delta E = \gamma \hbar (1 - \sigma) B_0 \quad (2.2)$$

with γ the gyromagnetic ratio (specific to each isotope), B_0 the applied magnetic field, and σ the chemical shielding around the nucleus. NMR transition frequencies depend on the distribution of electrons around the nucleus: they “shield” the latter from the applied magnetic field, accounted by σ in Equation (2.2). As resonance frequencies are function of the local environment, the NMR spectra obtained by Fourier transform will encode the chemical structure of the sample.^[6]

More on Spin

In quantum mechanics, spin is an intrinsic form of particle angular momentum. While being not related to a literal physical rotation, it gives rise to a magnetic moment, meaning the nucleus behaves like a tiny magnet. Only nuclei with a non-zero spin quantum number ($I \neq 0$) possess a net magnetic moment and therefore are NMR active. The most commonly used nucleus are ^1H , ^{13}C or ^{15}N .^[24]

As illustrated in Figure 2.3, the energy splitting arising from the interaction between nuclear spin and an external magnetic field leads to a characteristic resonance condition defined by the so-called “Larmor frequency”: $\omega_0 = \gamma B_0$. At resonance, an applied radio-frequency matching ω_0 can induce transitions between the spin states. This is the basis of NMR signal detection.^[25]

The Larmor frequency can be interpreted in terms of spin precession: the nuclear magnetic moment precesses around the B_0 axis at the angular frequency ω_0 , describing the same motion as a spinning top in a gravitational field. This picture is often useful for interpreting NMR phenomena.

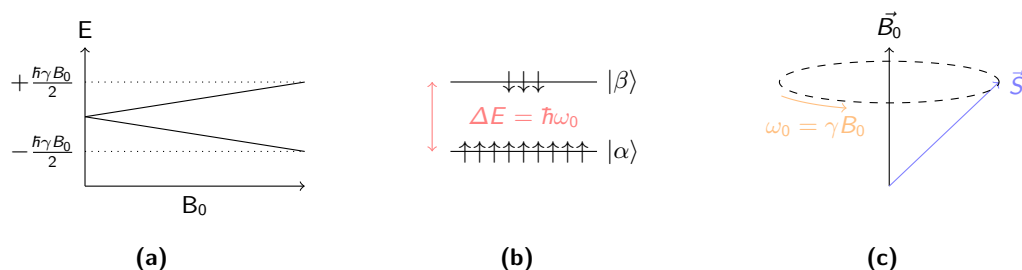


Figure 2.3 : $I = \frac{1}{2}$ nucleus in a static magnetic field B_0 : **(a)** Zeeman splitting into two levels corresponding to the spin projections $m = \pm \frac{1}{2}$. **(b)** Resulting thermal population distribution of the two states. **(c)** Spin precession at the Larmor frequency ω_0 , showing the rotation of the spin angular momentum $\vec{S}$ around the external magnetic field $\vec{B}_0$.

Magnetization and Polarisation

The total magnetic moment of a sample is referred to as the macroscopic magnetization M . It arises from the population distribution of nuclear spins among the energy levels. At thermal equilibrium, these populations follow the Boltzmann distribution, with a slight excess in the lowest energy state.^[25] Using thermostatic tools, this imbalance is often defined by the spin polarization P , calculated here for a $I = \frac{1}{2}$ nucleus:

$$P \equiv \frac{N_{\downarrow} - N_{\uparrow}}{N_{\downarrow} + N_{\uparrow}} = \tanh\left(\frac{\gamma\hbar B_0}{2k_B T}\right) \approx \frac{\gamma\hbar B_0}{2k_B T} \quad (2.4)$$

For protons (^1H) at room temperature and magnetic fields of a few tesla, the resulting polarization is very small, around 1/10000.^[6] This shows the intrinsically weak signal in NMR, although it is still sufficient to produce a measurable magnetization M . NMR signal's intensity is proportional to this magnetization :

$$M = C_p \cdot N \cdot P \quad \Rightarrow \quad M \propto P \propto \frac{\gamma B_0}{T} \quad (2.5)$$

with C_p the molar spin capacity (the energy of one mole of spins aligned to the same direction) and N the total number of spin. From this relation, we get an overview the strategies that can be considered to improve the signal–noise ratio. One could try increasing the magnetic field B_0 (reaching 1.2 GHz proton Larmor frequency, corresponding to about 28.2 T) or lower the temperature T (down to 30/40 K with helium cryoprobes). Unfortunately both of those strategies are experimentally and financially demanding.^[21]

As illustrated in Figure 2.6, an alternative route to reach “hyperpolarized” states will be playing on the gyromagnetic ratio γ , particularly by transferring polarization from electron spins *via* Dynamic Nuclear Polarization (DNP).

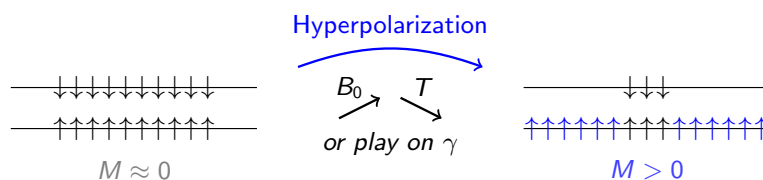


Figure 2.6 : From low polarization at thermal equilibrium to hyperpolarized states, resulting in improved NMR signals. The spins that contribute to magnetization are marked in blue.

2.3 Dynamic Nuclear Polarization (DNP)

Definition. Dynamic Nuclear Polarization (DNP) is a technique that enhances nuclear polarization by transferring spin polarization from unpaired electron spins, typically belonging to a polarizing agent.

The basic idea of DNP in solid-state NMR under MAS is illustrated in Figure 2.8. A polarizing agent (like binitroxide for instance), which provides unpaired electrons, is usually dissolved in a solution that forms a glass. When the sample is irradiated with microwaves close to the right EPR frequency, the electron spins become partially

saturated and their polarization is then transferred to nearby nuclear spins via magnetic coupling. A detailed description of the underlying mechanisms is beyond the scope of this report, but the reader may refer to [7] [8] for further understanding.

For example, electron/proton gyromagnetic ratio is approximately $\gamma_e/\gamma_H \approx 660$. Thus DNP can increase signal intensity by several hundred-fold, leading to even greater time gain for achieving a given signal-to-noise ratio.

Using the advantageous gyromagnetic ratio between nucleus and electron to transfer polarization was proposed quite early by Overhauser in 1953 but the implementation of DNP took several decades to become experimentally possible at high magnetic field. [26] However, since the 1990s, and in particular thanks to the development of high-power microwave sources by Griffin's group at MIT, DNP has been developed a lot and applied to a wide range of systems, from biomolecules to materials. [27]

Enhancement Factors and Build-up Times

To quantify the sensitivity achieved by applying DNP, the so called “enhancement factor” ε is defined as the ratio of the NMR signal intensity with microwave irradiation (μw ON) to that without irradiation (μw OFF), while the time-dependent signal under irradiation $S_{ON}(t)$ is commonly described by a mono-exponential build-up towards a steady state:

$$\varepsilon = \frac{I_{\mu w \text{ ON}}}{I_{\mu w \text{ OFF}}} \quad S_{ON}(t) = S_0 \left[1 - \exp\left(-\frac{t}{T_B}\right) \right] \quad (2.7)$$

This allows to introduce the other important observable : the so-called DNP “build-up time” T_B which characterizes how quickly the signal approaches its steady-state value under microwave irradiation.

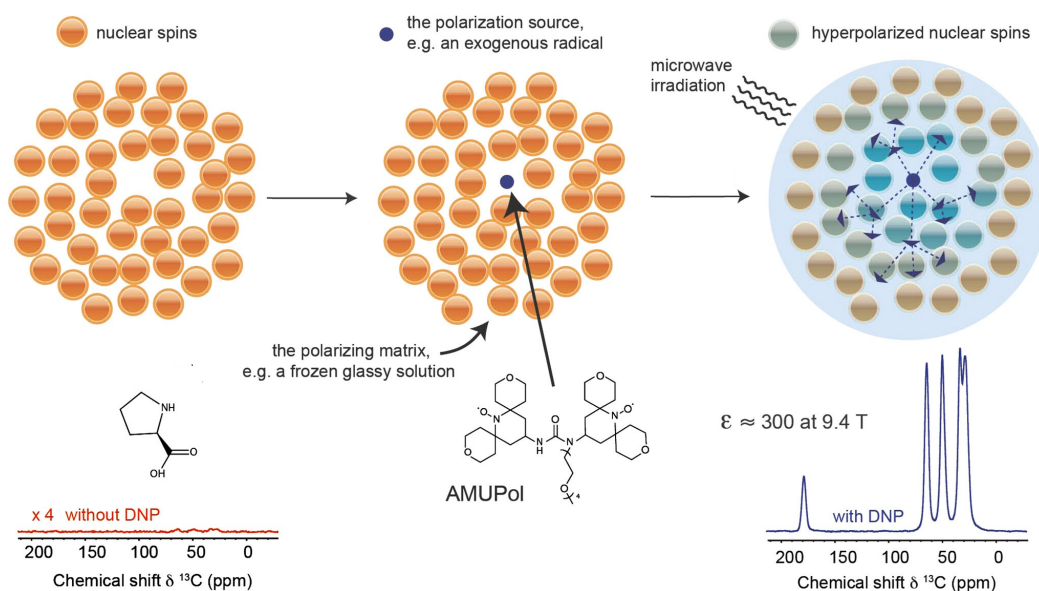


Figure 2.8 : Principle of DNP-enhanced solid-state MAS NMR spectroscopy, here illustrated for ^{13}C -labeled proline in frozen solution containing AMUPol binitroxide radical (chemical structure is shown in the figure) as the source of unpaired electrons.

Figure adapted from [7]

Spin Diffusion

SPIN diffusion plays a central role in hyperpolarization experiments. In order to transfer polarization from the radical to the nucleus of interest, we use proton spin diffusion: it's the Relayed Dynamic Nuclear Polarisation technique (R-DNP).

In this chapter, we will briefly explain how R-DNP can be used to probe the internal structure of LNPs, by monitoring how polarization propagates through the sample *via* proton spin diffusion.

3.1 Physical Mechanism

Definition. Spin diffusion is a process by which magnetization can be exchanged spontaneously between spins.

Although the nuclei themselves do not move, an effective transport of magnetization across the sample is observed, by dipolar interaction called “flip-flop transitions”, in which two neighbor spins exchange their spin states while conserving total magnetization.^[28] Analogously to thermal diffusion, polarization flows from highly polarized regions toward less polarized regions until a homogeneous distribution is reached.

Spin diffusion is characterized by the diffusion coefficient D , which describes how fast polarization propagates through the sample. In the rigid limit, it can be approximated as:

$$D \propto \gamma^2 c^{1/3} \quad (3.1)$$

where γ is the gyromagnetic ratio and c is the concentration of the nuclei.^[29] As a result, spin diffusion is more efficient in proton-rich samples and becomes less effective for nuclei with lower γ or concentration. In order to continue the transfer of polarization, an additional step is required, namely cross-polarization (CP). Unlike spin diffusion which is spontaneous, CP is an active polarization transfer achieved by a specific pulse sequence, presented in [Appendix B](#).

Thus, spin diffusion controls how polarization spreads in LNPs: component at the particle surface usually show higher enhancement factors and reach steady state faster than those in the particle core.^[30] As a result, enhancement factors and build-up times can provide useful indirect information on the internal architecture of the LNPs. However, each component has to have a clearly resolvable resonance in the NMR spectrum.

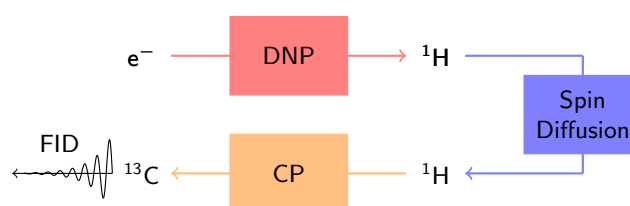


Figure 3.2 : Schematic representation of the polarization transfer pathway in R-DNP.

3.2 Numerical simulation

To complement NMR measurements, numerical simulations are frequently employed to support the interpretation of experimental data. Even if spin diffusion can be described using quantum mechanical methods, these approaches become computationally expensive at the μm scale, as they require explicitly modeling the interactions between millions of individual spins. For this reason, a thermodynamic description is typically adopted, in which the nuclear magnetization evolves according to a diffusion equation. [28]

In this work, we use a numerical simulation of spin diffusion originally developed by A. Pinon in his thesis [29] and adapted to LNPs at CRMN by A. Rancz [10].

Using Equation (2.5), spin magnetization per unit of volume is write as :

$$M(x, t) = C_p C(x) P(x, t) \quad (3.3)$$

Its time evolution is described by a diffusion–relaxation equation that accounts for both spatial transport of magnetization local return to thermal equilibrium:

$$\underbrace{\frac{\partial M(x, t)}{\partial t}}_{\text{Temporal evolution}} = \underbrace{\nabla \cdot (D(x) \nabla P(x, t))}_{\text{Diffusion}} - \underbrace{\frac{M(x, t) - M_0(x, t)}{T_1(x)}}_{\text{Relaxation toward equilibrium}} \quad (3.4)$$

with $M(x, t)$ the local spin polarization, $D(x)$ the position–dependent spin diffusion coefficient, $M_0(x, t)$ the equilibrium magnetization toward which the system relaxes and $T_1(x)$ the longitudinal relaxation time, meaning the characteristic time describing how quickly the net magnetization recovers along the direction of the external magnetic field after being perturbed (T_B in DNP condition).

The resulting partial differential equation (PDE) is solved numerically in **MATLAB**, for the two different experimental conditions: μw OFF case, and μw ON case (DNP conditions). Once the PDE is solved, $M_{\text{ON/OFF}}(x, t)$ is integrated to obtain the total observable NMR signal $S_{\text{ON/OFF}}(t)$:

$$S_{\text{ON/OFF}}(t) = \frac{1}{V} \int_V M_{\text{ON/OFF}}(x, t) \theta(x) |J(x)| dx \quad (3.5)$$

with V is the integration volume, $\theta(x)$ a spatial quenching factor accounting for local signal attenuation effects and $J(x)$ the spherical Jacobian. Further information on the simulation including **MATLAB** code are given in Appendix C.

The system is modeled as a sphere with a radius corresponding to the typical size of a LNP. Protons are assumed to be continuously distributed throughout the domain, allowing spin transport to be described as isotropic diffusion across the entire system. The nanoparticle is represented using concentric shells, each associated with different LNP components. By integrating over the chosen domain (meaning the right shell), we are able to compute ε and fit T_B as defined in Equation (2.7) for each component.

Therefore, those numerical results can be used to help the interpretation of experimental data for any LNPs studied by solid–state NMR DNP under MAS conditions.

Part II

Results and Discussion

4 Experimental Results

THIS chapter presents the experimental work conducted during this internship, aimed at investigating the internal architecture of mRNA-loaded lipid nanoparticles, using NMR enhanced by DNP approach. It first describes the materials and experimental methods used throughout the study, before presenting the NMR results and their analysis.

4.1 Material and Methods

mRNA-loaded Lipid Nanoparticles

The mRNA-loaded lipid nanoparticles (mRNA-LNPs) were formulated by M. Brunetto from the group of G. Lollo (CNRS, LAGEPP, Université Claude Bernard Lyon 1) on June 22nd, using isotopically labelled mRNA provided by L. Gayet (CRMN). The formulated samples were transferred immediately to CRMN, where the experiments were performed during the following week.

mRNA-LNPs Formulation

mRNA-LNPs were formulated using a one-step microfluidic formulation method with a NanoAssemblr Ignite system. The organic phase was prepared by dissolving the lipids in ethanol at a total lipid concentration of 20 mM. The aqueous phase consisted of mRNA diluted in sodium acetate buffer (25 mM, pH 4.5). The formulation was performed using a 3:1 aqueous-to-organic flow rate ratio at a total flow rate of 12 mL/min.

Following formulation, the mRNA-LNPs suspension was purified by dialysis against sodium acetate buffer (25 mM, pH 4.5). Final composition is detailed in [Table 4.1](#) with approximately 5.96 mg/mL total lipids and 0.72 mg/mL mRNA.

DNP Sample Preparation

The sample was prepared with a 30% (V/V) solution of 36 mM of polarizing agent AMUPol (36 mM in ^{12}C -glycerol- d_8). A volume of 25 μL of this sample solution was transferred using a micropipette into a 3.2 mm sapphire rotor that was capped with a teflon plug.

DNP Experiments

DNP experiments were performed on a Bruker 400 MHz DNP NMR spectrometer (9.4 T) equipped with a gyrotron system (μw source) and a 3.2 mm MAS probe. Measurements were carried out under MAS conditions at a spinning frequency of 12.5 kHz and a sample temperature of approximately 100 K. μw irradiation was applied at 263 GHz, corresponding to the electron resonance frequency of the gyrotron system. More detailed information about instrumentation are provided in [Appendix B](#).

^{13}C cross-polarization (CP) experiments were performed under both DNP ON and DNP OFF conditions, and the resulting spectra were compared to evaluate the polarization

enhancement. A contact time of 1.5 ms and a recycle delay of 2.6 s were used. Proton heteronuclear decoupling was applied during acquisition with a radiofrequency field strength of 50 kHz.

DNP build-up curves were acquired using ^{13}C CP saturation recovery experiments (μw ON) with recovery delays ranging from 0.01 to 40 s (10 logarithmically spaced delay times). The same parameters as the ones used for the ^{13}C CP experiments are applied. More detailed information about NMR sequences can be found in [Appendix B](#).

Concentration (mg/ml)			
C12-200	DOPE	DOTAP	mRNA
3.991 (37.5%)	0.834 (25%)	1.132 (37.5%)	0.720

Table 4.1 : Composition of the mRNA-LNPs formulation, lipid molar fractions are indicated in gray. Chemical structures are presented in [Appendix A](#).

Pure Lipids

Pure lipid samples were prepared as reference compounds for spectral assignment and were provided by LAGEPP (10 mg of each lipid).

Sample Preparation

DOPE was packed in a 2.5 mm MAS rotor. DOTAP and C12-200 were provided in a pasty oil-like form, hence liquid-state NMR was required. The latter were dissolved directly in deuterated chloroform CDCl_3 ($\approx 170\mu\text{L}$). A volume of 150 μL of this sample was transferred using a micropipette into 3 mm NMR tubes for analysis.

NMR Experiments

Solid-state NMR measurements of DOPE were performed on a Bruker 500 MHz NMR spectrometer (11.7 T) equipped with a 2.5 mm MAS probe. ^{13}C CP experiments were conducted at a spinning frequency of 20 kHz and maintained at 285 K to preserve the solid-state nature of the sample.

Liquid-state NMR experiments were performed on a Bruker 600 MHz NMR spectrometer (14.1 T). Direct ^{13}C detection with broadband proton decoupling were conducted using standard `zgpg30` experiment.

Number of scans		
mRNA-loaded Lipid Nanoparticles		
^{13}C CP – μw ON	^{13}C CP – μw OFF	^{13}C Sartrec – μw ON
6376 ($\approx 4\text{h}30$)	57344 ($\approx 41\text{h}45$)	13440 ($\approx 31\text{h}45$)
DOPE	C12-200	DOTAP
^{13}C CP	^{13}C <code>zgpg30</code>	^{13}C <code>zgpg30</code>
24576 ($\approx 13\text{h}45$)	2048 ($\approx 1\text{h}30$)	316 ($\approx 15\text{min}$)

Table 4.2 : Number of scans used for the solid-state and liquid-state NMR measurements. Approximate acquisition times are indicated in gray.

4.2 Experimental Data Analysis

Spectral Assignment

The main spectra of the mRNA–LNPs sample under DNP condition was assigned by comparison with pure lipid samples reference spectra and previous assignments from A. Rancz’s thesis [10]. This assignment was challenging due to the presence of intense residual ethanol signals, mainly observed around 60 and 20 ppm. These signals strongly overlap with several lipid resonances, complicating the identification of individual components.

In the spectra of the pure lipid references, characteristic resonances can be identified: the carboxylic acid resonance of CDCl_3 , appearing around 72 ppm, visible in the liquid-state for C12-200 and DOTAP spectra and the $\text{C}=\text{C}$ resonance of unsaturated lipid chains, expected around 130 ppm, for the DOPE and DOTAP spectra. However, the latter resonance was not clearly observed in the mRNA–LNPs spectra, even in DNP condition. This should be related to the much lower lipid concentration used in this study (6 mg/mL vs 25 mg/mL in Rancz’s thesis). Only a resonance around 35 ppm (1a) could be assigned to CH_2 groups from lipid components, most likely dominated by C12-200 due to its higher molar fraction within the LNP formulation.

In A. Rancz’s thesis, where the mRNA was not isotopically labelled, only a limited number of mRNA resonances could be detected. Among them, a resonance around 90 ppm was assigned to the C1' (sugar) carbon of the mRNA backbone, which is also observed in the present spectra. Additionally, three other resonances (1b), (2b) and (3b) are tentatively attributed to mRNA, as they are not observed in any of the lipid reference spectra and cannot be assigned to spinning sidebands. This assignment is supported for (2b) and (3b) by a ^{13}C chemical shift distributions reported in the nucleic acid histograms of the Biological Magnetic Resonance Bank (BMRB). [31]

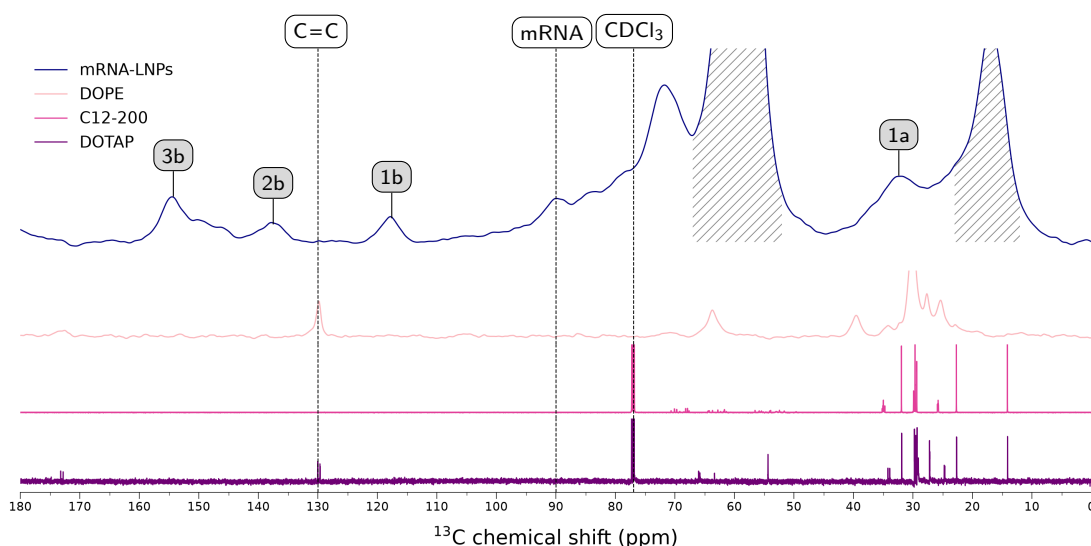


Figure 4.3 : Proposed spectral assignment. From top to bottom: mRNA–LNPs under DNP condition, DOPE, C12-200, and DOTAP reference spectra. The two hatched peaks correspond to residual ethanol signals. (1a) is assigned to lipid components, while (1b), (2b), and (3b) are likely associated with mRNA.

General Results

Ideally, spectral deconvolution should be used to separate overlapping resonances and obtain more reliable quantitative information. Although some attempts were made, time constraints and inconclusive results led to the use of simpler extraction methods in the present analysis. Two approaches were considered: (i) automatic peak picking based on the maximum signal intensity, and (ii) integration over a defined spectral window centered on the detected peak position.

Enhancement Factors

For each signal of interest, the resonance position was automatically refined by searching for the maximum intensity compared to the given reference within a ± 0.5 ppm window. The peak height at this position was then used to determine the enhancement factor, as defined in Equation (2.7). In addition, integrated peaks were calculated using symmetric integration windows of increasing width (0.02 – 2 ppm) centered on the detected peak position, in order to calculate an average ε value. Results are presented in Figure 4.4.

mRNA is expected to have a lower enhancement since it is located in the core region of the LNPs, where polarization transfer is less efficient (greater distance from the polarizing agents). In contrast, lipid components located in the outer shell are expected to experience more efficient polarization transfer and therefore higher enhancement.

This trend is observed for (3b), which has an $\varepsilon \approx 53$, while the resonance attributed to lipid components (1a) shows a higher enhancement factor of $\varepsilon \approx 63$. However, the trend is less pronounced for (2b) and is clearly not followed by (1b), which have both a higher enhancement and a larger deviation. This could suggest that a fraction of the mRNA is closer to the particle surface, potentially interacting with ionizable or cationic lipids. Nevertheless, mRNA resonances would be expected to have comparable enhancement factors, which is not the case here: the signal-to-noise ratio is not sufficient to ensure a consistent analysis.

This is illustrated in Figure 4.5, which shows the dependence of ε on the integration window. The most reliable integration-based ε values were obtained for (1a) and (3b), for which the expected trend is observed.

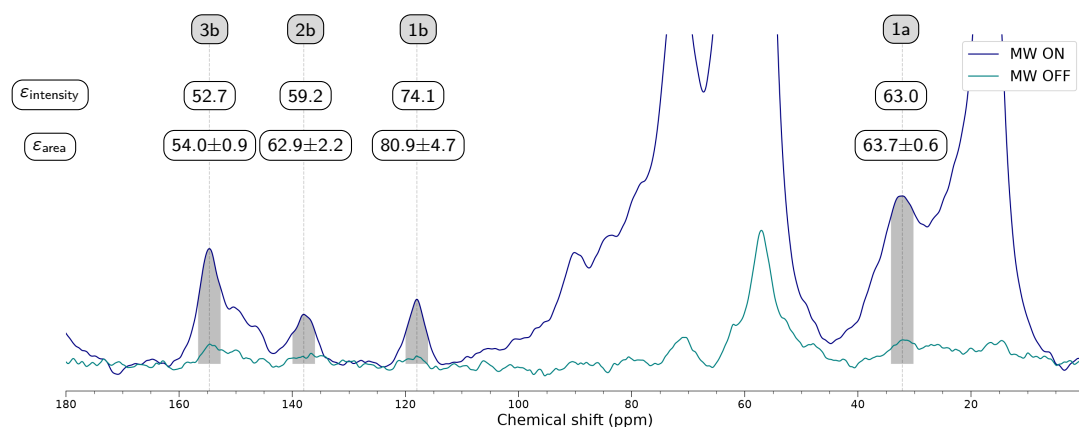


Figure 4.4 : DNP enhancement factor (ε) obtained for the different resonances. Gray shaded areas represent the maximum integration windows.

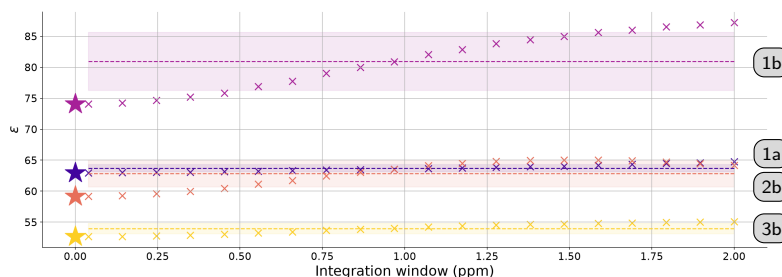


Figure 4.5 : DNP enhancement factor (ε) versus integration window. From top to bottom: (1b), (1a), (2b), and (3b). The dotted line and shaded area indicate respectively the mean ε value and standard deviation. The star denotes the intensity-derived ε . Resonances (1a) and (3b) show the highest stability.

Build-ups

The signal evolution was measured over ten experiments with varying recovery delays in CP saturation recovery experiments. The same approach as before was applied, for each resonance, the extracted intensities (or integrated areas) were fit using the exponential build-up model described in Equation (2.7), in order to extract the polarization build-up time T_B .

Molecules located in the core of the particle are expected to have larger T_B , as they are further from the polarizing agents and therefore require more time to reach the hyperpolarized state. Conversely, those near the particle surface should have shorter T_B . Consistently with the previous analysis, higher T_B values are therefore expected for mRNA resonances, while lipid components should have faster polarization build-up.

The mRNA resonance at 90 ppm was also included. Its assignment is consistent with the previous analysis, as this resonance and (1b) both have $T_B \approx 3$ s, compared to $T_B \approx 2.3$ s for the lipid resonance (1a). The lower T_B values observed for (2b) and (3b) don't follow this interpretation, which may arise from the limitations mentioned previously : insufficient lipid concentration, overlapping with residual ethanol... Future work should address those problems during sample preparation and use spectral deconvolution in order to improve the accuracy and reliability of the quantitative analysis.

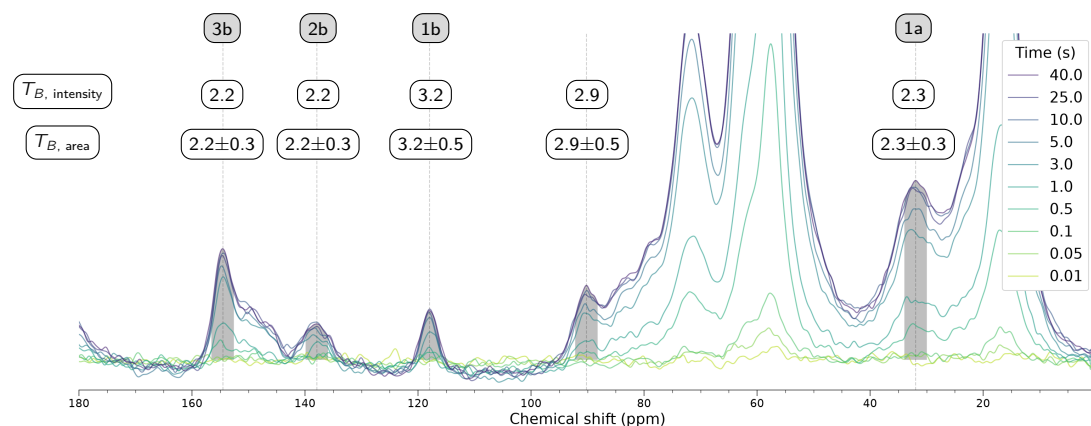


Figure 4.6 : Polarization build-up time (T_B) extracted for the different resonances. Gray shaded areas represent the maximum integration windows.

Numerical Calculations

THE numerical calculations presented in this chapter are based on the work of A. Rancz [10] at the CMRN. This internship was not aimed to modify the fundamental physical framework of the simulation, but rather to understand its implementation and extend its use toward the investigation of different LNP models. Using the `matlab.engine` library, the MATLAB spin diffusion simulation was integrated into a more flexible python-based workflow. MATLAB code is presented in Appendix C, while the complete python implementation will be made available in a GitHub repository.

This chapter briefly presents the fitting attempt of simulation data to the experimental results obtained during this internship, before extending the approach toward more complex structural models.

5.1 Comparison Results

Based on the LNP composition reported in Table 4.1, the estimated volume fractions of the different components are: 6% mRNA, 63% C12-200, 13% DOPE, and 18% DOTAP. Most of the simulation parameters were estimated in order to obtain a better agreement with the experimental results, more details can be found in Appendix C. The experimental values retained for comparison are: $\epsilon_{\text{mRNA}} = 53$, $\epsilon_{\text{lipid}} = 63$, $T_{B, \text{mRNA}} = 3.0$ s, and $T_{B, \text{lipid}} = 2.3$ s.

Four models are illustrated: (i) an homogeneous model; (ii) a model with mRNA located at the centre and lipids surrounding it; (iii) a model with an mRNA core, followed by an intermediate ionizable lipid layer and an outer lipid region; and finally (iv) an inverted model with the opposite spatial organization (included for illustrative purposes).

This work is presented to show the potential of the simulation but interpreting results remains limited due to the lack of reliability of the experimental data. Nevertheless, models (ii) and (iii) show encouraging agreement with the experimental data, notably in reproducing the enhancement factors ϵ for (ii) and the build-up times T_B for (iii).

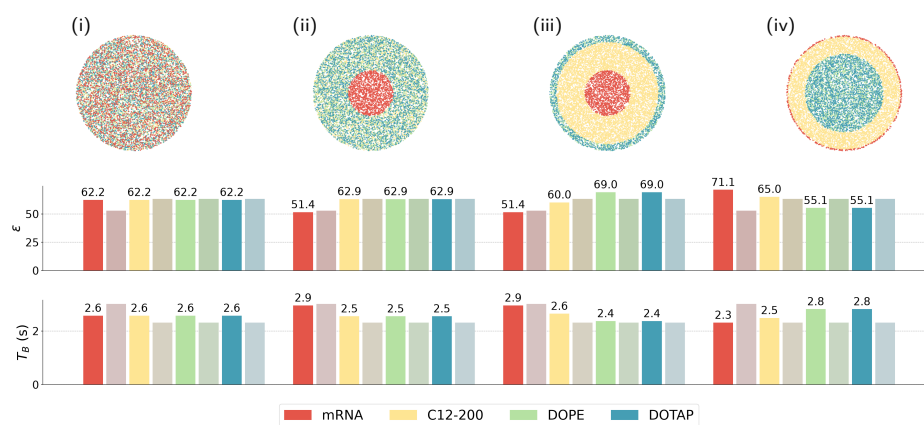


Figure 5.1 : Schematic representation of the LNP models considered and their observables: (i) homogeneous reference model; (ii) mRNA-core/lipid-shell, (iii) mRNA-ionizable lipid-outer lipid, and (vi) inverted model. Experimental values are shown in darker color for comparison.

5.2 Beyond Homogeneous LNP Models

By modifying the geometrical parameters of the system, it became possible to rapidly simulate different structural configurations and evaluate their impact on the DNP observables. The parameters used in the LNP models shown below are hypothetical, but they could be adapted and fitted to support the analysis of experimental data. The polarization build-up curves were fitted using stretched exponential functions, which provide a more robust description when integrating over a broader spectral domain, as detailed in [Appendix C](#).

Gradient Concentration

One possibility is to simulate heterogeneous lipid organization with concentration gradients inside the LNP. Instead of assigning a given component entirely to a single shell, its concentration could be distributed across several shells. The contribution of each shell is calculated, and the observables associated with this multi-component shell are obtained by weighting the contribution of each shell according to the desired spatial distribution.

Here, we simulate an LNP model containing mRNA, cholesterol, and three additional lipids (A, B, and C). The mRNA is always located in the core shell. Three structural models are illustrated: (i) a reference model with each component assigned to a separate shell; (ii) a model where cholesterol is homogeneously distributed within the other lipid shells; and finally (iii) a model with different cholesterol/lipid ratios of 2:1, 1:1, and 1:10. The impact of these concentration gradients on spin diffusion and polarization transport can be quantitatively assessed through the evolution of the simulated observables.

This has been further explored in A. Rancz's thesis, based on the models proposed in [\[32\]](#) [\[33\]](#), where cholesterol is suggested to be distributed across multiple lipid regions rather than confined to a single shell.

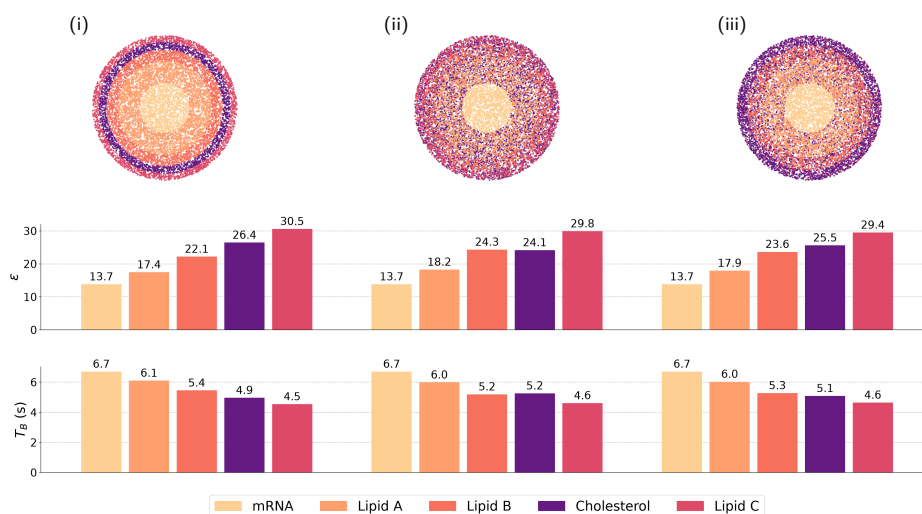


Figure 5.2 : Schematic representation of the LNP models considered and their observables: (i) reference model with each component assigned to a separate shell; (ii) with cholesterol homogeneously distributed within the lipid shells; (iii) with different cholesterol/lipid ratios (2:1, 1:1, and 1:10). The mRNA is located in the core shell for all models.

Internal mRNA Distribution

Another structural parameter of interest is the spatial distribution of mRNA within the LNP. Instead of assuming that mRNA is entirely confined to a single region, we investigate models where its concentration is progressively redistributed across different shells.

A reference model (i) is defined by assuming that 100% of the mRNA is located in the core region. Additional models progressively redistribute mRNA toward the surrounding shells, considering the following core/outer mRNA distributions: (ii) 1:4, (iii) 4:1, and (iv) where all the mRNA is in the outer shell.

The simulated observables associated with these two mRNA “types” (namely “core” and “outer” mRNA) can be compared through the simulations. In fact, DNP observables extracted from experimental data would correspond to an average behaviour. The corresponding average signal is also calculated, and the resulting DNP observables are labelled as “Total mRNA” in the figure. Only mRNA results are presented here, but the same analysis could be easily performed for lipid.

Overall, these simulations confirm that mRNA located closer to the particle surface experiences faster and more efficient polarization transfer. The enhancement is increased by approximately a factor of three when comparing mRNA located in the outer shell to mRNA confined in the core region and the polarization build-up times decrease from approximately 6.7 s in the core to 4.4 s in the outer shell, indicating a significantly faster polarization transfer process for mRNA outside. These simulations show also that DNP observables can be used to discriminate whether LNPs contain mRNA confined exclusively to the core or partially into the outer shell.

This approach may be relevant for the “two-step” formulation strategy proposed by the LAGEPP team (G. Lollo), which collaborates with CMRN for the LNP project. In this novel formulation, mRNA is suggested to be at least both located closer to the particle surface and in the inner core.

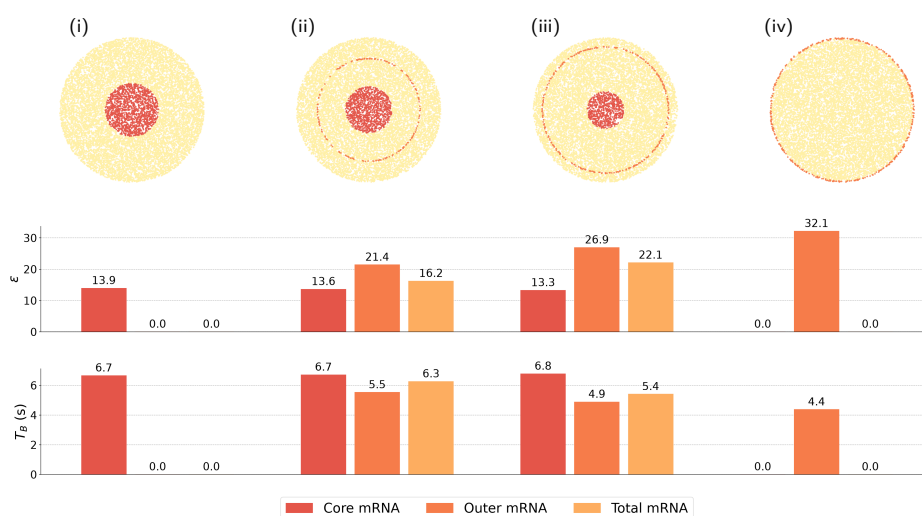


Figure 5.3 : Schematic representation of the LNP models, varying mRNA spatial distributions and their observables : (i) 100% mRNA in the core, (ii) core/outer mRNA ratio of 1:4, (iii) ratio of 4:1, and (iv) 100% mRNA in the outer shell.

Asymmetric Structures

A further extension of the model is to simulate asymmetric structures, such as half-shell models, in which different compositions are assigned to distinct spatial regions of the nanoparticle.

Although the model relies on spherical symmetry and cannot explicitly describe angularly asymmetric structures, such configurations can still be simulated as spin diffusion is homogeneous throughout the LNP in our model (the proton concentration is considered constant). Therefore, an asymmetric composition can be represented by assigning mRNA and a corresponding amount of lipid to the core region, while distributing the remaining lipid composition in the surrounding shell. The resulting mRNA DNP observables are then calculated by considering only half of the signal, according to the volume fraction occupied by the latter in the core.

Here, we simulate a reference model and the corresponding asymmetric half-shell model, in which all the mRNA is confined to one half of the core shell. The simulations show that such an asymmetric architecture induces only a little change in ε and an almost negligible change in T_B relative to the spherical model.

Such an approach could be relevant for describing LNPs exhibiting non-uniform internal organizations. For example, in [34], self-amplifying mRNA (SAM) LNPs were reported to exhibit characteristic “bleb” structures, consisting of a lipid-rich core and an aqueous RNA-rich core surrounded by a rich lipid shell. These structural features were characterized using SANS, SAXS, and cryo-EM measurements.

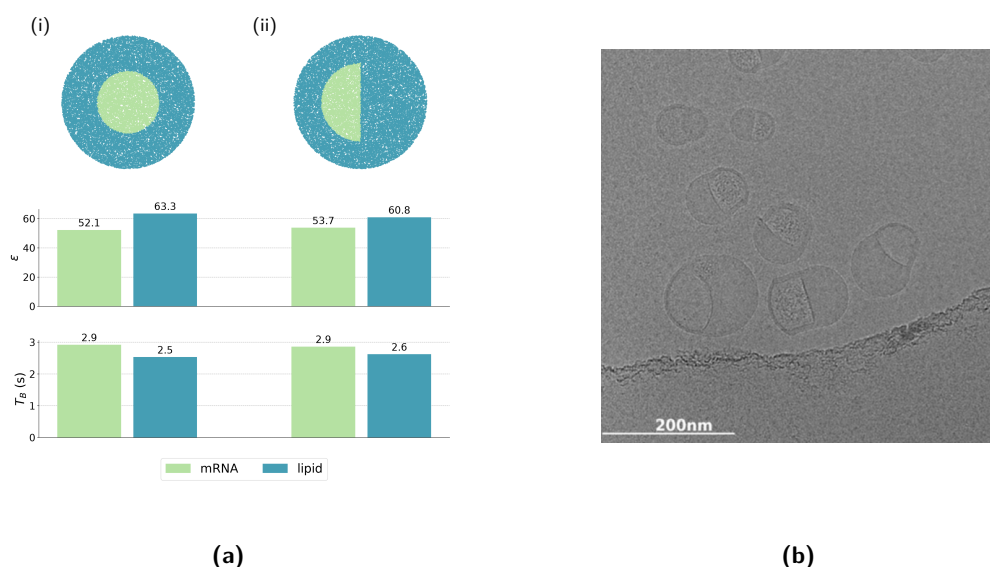


Figure 5.4 : (a) Schematic representation of the LNP half-shell models and their corresponding observables: (i) reference model; (ii) half-shell model. (b) Cryo-EM image of SAM LNPs, indicating that a two-compartment (“bleb”) structure may be the predominant morphology. The clear compartment is the mRNA-rich region.

(b) from [34]

Conclusion

THIS two-month internship was aiming to contribute to the structural characterization of a new formulation of lipid nanoparticles developed within a collaboration between the CRMN (A. Lesage group) and the LAGEPP (G. Lollo group). This work is the first set of experiment within this new collaboration, with the objective of gaining further insight into the molecular organization of lipid nanoparticles designed for RNA delivery applications. In continuation of A. Rancz's PhD, this internship contributed to the improvement of a proton spin diffusion simulation approach. By modeling the transfer of polarization through the proton network, these simulations provide a complementary method to interpret experimental DNP-NMR results.

This double approach, combining simulations and experiments, remains strongly dependent on both the accuracy of the models used and the quality of the experimental data. The preparation of samples with higher LNPs concentrations will represent an important step toward overcoming the low signal-to-noise ratio leading to unreliable results. However, achieving this could be challenging for the formulation team, as it must be performed without compromising the physicochemical properties and structural organization of the LNPs. An additional strategy could be to run experiments on LNPs loaded with unlabelled mRNA in order to see more easily lipid signals and facilitate their interpretation. Together with the complete removal of residual ethanol and more reliable spectral deconvolution, these approaches would enable a more accurate determination of DNP observables.

More broadly, this work highlights the potential of combining solid-state NMR under MAS, DNP techniques, and computational modeling to investigate complex nanostructured systems. A deeper understanding of the molecular organization of lipid nanoparticles would be helpful to improve their formulation and optimizing their ability to deliver RNA medicines. In the longer term, the combination of experimental characterization and predictive simulations could contribute to the development of more rational strategies for the design of next-generation lipid nanoparticle-based delivery systems.

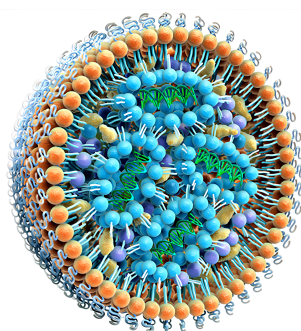
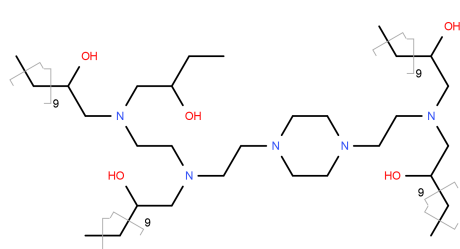


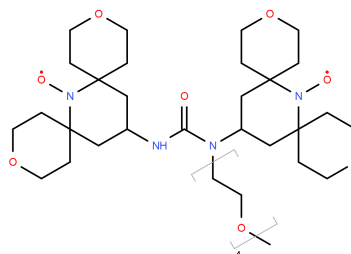
Figure 5.5 : Schematic cross-section of a lipid nanoparticle, with encapsulated mRNA (in green) at the core and several different lipid species distributed around it.

Appendix

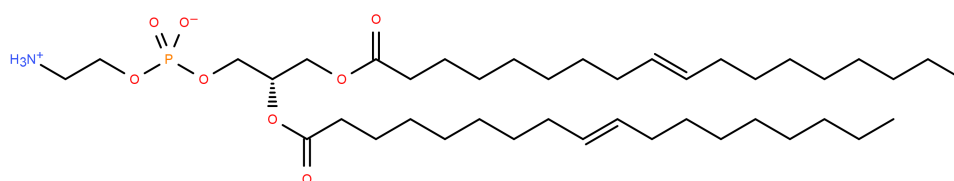
Chemical Structures



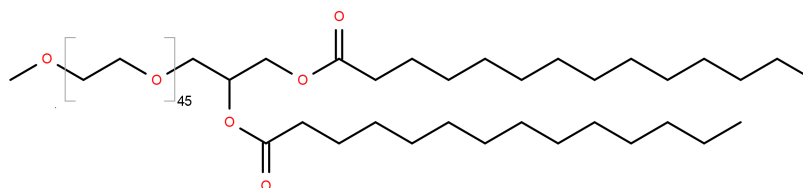
(a) Ionizable lipid C12-200



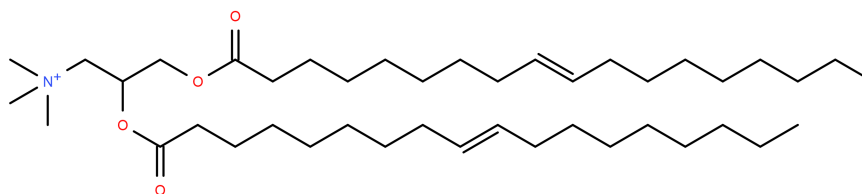
(b) Polarizing agent AMUPol



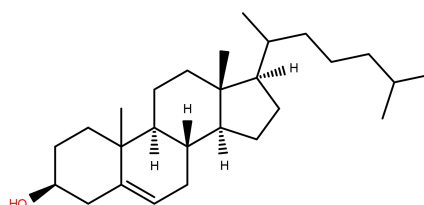
(c) Phospholipid DOPE



(d) PEGylated lipid PEG-DMG



(e) Cationic lipid DOTAP



(f) Cholesterol

Figure A.1 : Chemical structures of the molecules presented in this study: ionizable lipid C12-200 (a), polarizing agent AMUPol (b), and helper lipid (c–f).

NMR Experimental Setup

B.1 Cross Polarization

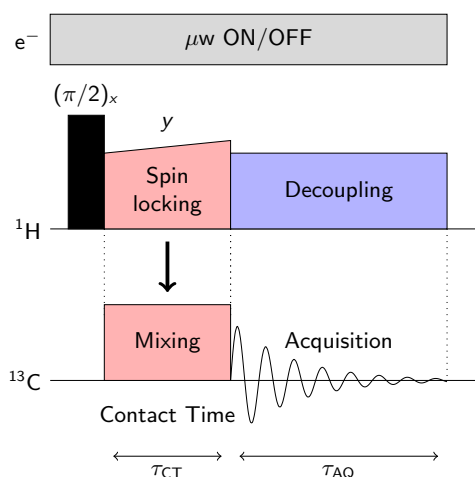


Figure B.1 : Pulse sequence for cross-polarization (CP) in solid-state NMR. Under DNP conditions, continuous-wave (μw ON) irradiation is applied during the entire experiment. A 90° pulse is applied to the ^1H spins, followed by polarization transfer to dilute nuclei (^{13}C here) during the contact time τ_{CT} under the Hartmann–Hahn matching condition, where the radiofrequency fields applied to both nuclei are adjusted to equalize their effective nutation frequencies. The NMR signal is acquired over the acquisition time τ_{AQ} while applying continuous ^1H decoupling.

B.2 Saturation Recovery Experiments

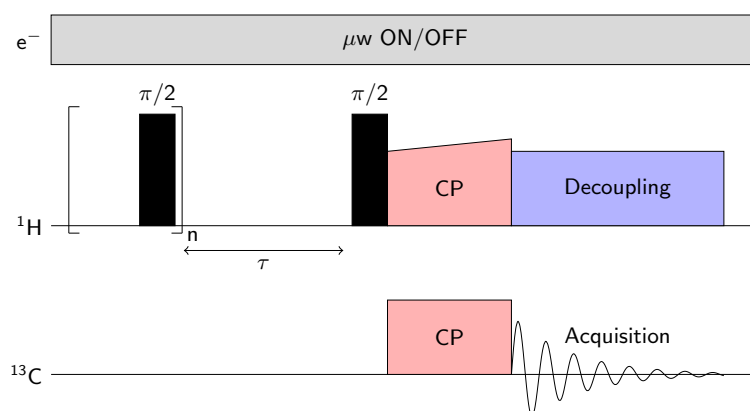


Figure B.2 : Pulse sequence for ^{13}C CP saturation-recovery experiment in solid-state NMR. Under DNP conditions, continuous-wave (μw ON) irradiation is applied during the entire experiment. A ^1H 90° pulse is applied to remove transverse ^1H magnetization. Then, during a variable delay τ , the polarization is transferred from the electrons to nearby protons via spin diffusion. Polarization is then transferred to ^{13}C via CP and detected. This provides kinetic information on spin diffusion throughout the sample, enabling here the determination of the DNP build-up time.

B.3 Instrumentation

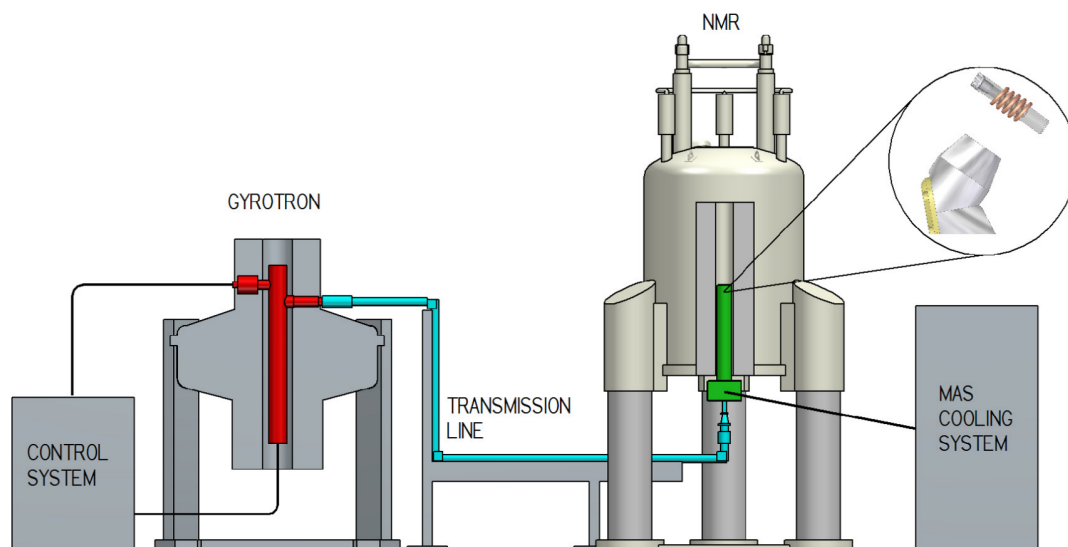


Figure B.3 : Schematic representation of a conventional DNP MAS NMR setup. The gyrotron, located on the left, is used as the μw source, and the latter generated are transferred through a waveguide to the low-temperature MAS probe positioned inside the NMR magnet. The gases used for rotor spinning are cooled using the MAS cooling system.

Figure adapted from [35]

Numerical Implementation

BASED on the LNP composition, the volume fractions of the different components can be calculated. These fractions are then used to construct the structural model of the LNP by defining shell radii. Spin diffusion is simulated, and the resulting signal contribution is integrated according to Equation (3.5). This approach enables the extraction of the simulated DNP enhancements and polarization build-up times.

C.1 Parameters

The main parameters used for the simulations are summarized in Figure C.1. The model distinguishes two main regions: the *source* region, corresponding to the area containing the polarizing agent where the initial hyperpolarization is generated, and the *target* region, corresponding to the surrounding LNP material where polarization is transferred through spin diffusion.

L_t and L_s (target and source lengths) : define the geometrical dimensions of the simulated regions and determine the distance over which polarization must diffuse.

ε_0 (source enhancement factor) : describes the initial enhancement of the nuclear polarization in the source phase produced by DNP. It defines the maximum polarization gain that can be transferred to the target phase.

$T_{B,s}$ (source build-up time) : corresponds to the characteristic time required for the polarization in the source phase to reach its steady-state value under microwave irradiation.

$T_{1,t}$ (target longitudinal relaxation time) : describes the time constant governing the return of the target polarization toward thermal equilibrium after polarization transfer.

$c_{1H,t}$ and $c_{1H,s}$ (proton concentrations) : represent the local density of protons available for polarization transfer in each region.

D_t and D_s , (diffusion coefficients) : describe how rapidly polarization spreads through each phase by spin diffusion.

In addition, a hyperbolic tangent function is used to describe the spatial transition of T_1 , D , and c_{1H} at the interface between the source and target phases.

Together, these parameters define the initial polarization conditions, relaxation behavior, spin diffusion properties, and geometry required to simulate the experimental polarization transfer dynamics. Their values are obtained from experimental measurements, estimated when necessary, or extracted from the literature. Interested readers may refer to A. Rancz's thesis.

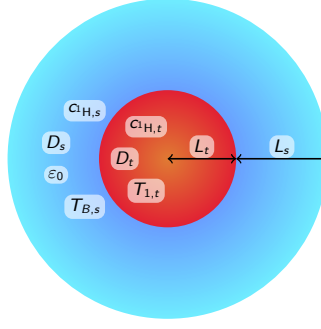


Figure C.1 : Schematic representation of the source–target model used for the spin diffusion simulations. The source region in blue (length L_s) contains the polarizing agent and is characterized by the DNP enhancement factor ϵ_0 , build–up time $T_{B,s}$, proton concentration $c_{H,s}$, and spin diffusion coefficient D_s . Polarization is transferred by spin diffusion to the target region in red (length L_t), which is described by the longitudinal relaxation time $T_{1,t}$, proton concentration $c_{H,t}$, and diffusion coefficient D_t .

C.2 Polarization build–up fitting

The polarization build–up curves were analyzed using exponential models. A mono-exponential function, defined in Equation (2.7), assumes a single characteristic build–up time. Because the signals were obtained by integrating over relatively broad spectral regions, the measured build–up curves may contain contributions from multiple environments with different polarization transfer efficiencies.

In this case, a stretched exponential model provides a more appropriate description by accounting for a distribution of characteristic time constants. The polarization build–up is then expressed as :

$$P(t) = 1 - \exp \left[- \left(\frac{t}{T_B} \right)^\beta \right] \quad (\text{C.2})$$

where T_B represents the characteristic build–up time and β is the stretching parameter. A comparison between the mono-exponential and stretched exponential for the homogeneous model shown in Figure 5.1 fits is shown below.

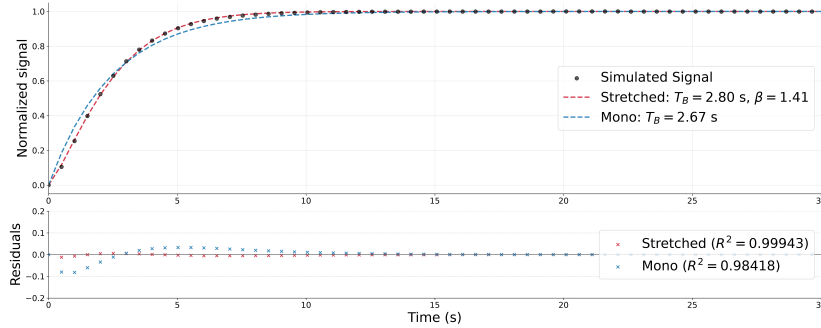


Figure C.3 : Normalized polarization build–up curve for the homogeneous model shown in Figure 5.1, fitted using mono-exponential and stretched exponential models. The stretched exponential reproduces the experimental data more accurately, particularly at short polarization times, as evidenced by the residuals. The fitted characteristic build–up times differ by approximately 0.2 s here.

C.3 Matlab Code

```

1  function signals = spin_diffusion(m, length_target, rstep, tmax, tstep, Tlout, Tlin, epsilon0,
   ↪ Din, Dout, length_source, slope, theta, dep, concjuice, concX, Cp, radii)
2
3      % Spatial and temporal discretization
4      r = linspace(0, length_target + length_source, rstep);
5      t = linspace(0, tmax, tstep);
6
7      % Solve spin-diffusion equation with microwave irradiation (DNP ON)
8      sol = pdepe(m, @pdex1pde, @pdex1ic, @pdex1bc, r, t);
9      p = sol(:,:,1);
10
11     % Solve the diffusion equation without microwave irradiation (DNP OFF)
12     sol2 = pdepe(m, @pdex1pdenoDNP, @pdex1icnoDNP, @pdex1bcnoDNP, r, t);
13     pref = sol2(:,:,1);
14
15     % Keep only the polarization that contributes to the detected NMR signal.
16     p_cut = zeros(length(t), length(r));
17     pref_cut = zeros(length(t), length(r));
18
19     for l = 1:length(r)
20         p_cut(:,l) = p(:,l) * theta0(r(l)); % DNP ON
21         pref_cut(:,l) = pref(:,l) * theta0(r(l)); % DNP OFF
22     end
23
24     % Number of concentric shells defined by the input radii
25     n_shells = length(radii) - 1;
26     signals = struct('S_off', {}, 'S_on', {}, 't', {});
27
28     % Compute the signal from each shell
29     for i = 1:n_shells
30
31         % Radial limits of the current shell
32         r_min = radii(i);
33         r_max = radii(i + 1);
34
35         i_min = find(r >= r_min, 1, 'first');
36         i_max = find(r <= r_max, 1, 'last');
37
38         Presign = zeros(length(t), length(r));
39         Presignref = zeros(length(t), length(r));
40         SignalPol = zeros(1, length(t));
41         SignalRef = zeros(1, length(t));
42
43         % Compute the pre-signal
44         for j = 1:length(t)
45             for k = i_min:i_max
46                 Presign(j,k) = fc(r(k)) * p_cut(j,k) * r(k)^m;
47                 Presignref(j,k) = fc(r(k)) * pref_cut(j,k) * r(k)^m;
48             end
49         end
50
51         %% Integrate the pre-signal over the shell
52         for w = 1:length(t)
53             SignalPol(w) = trapz(Presign(w,:));
54             SignalRef(w) = trapz(Presignref(w,:));
55         end
56
57         signals.S_off{i} = SignalRef;
58         signals.S_on{i} = SignalPol;
59         signals.t{i} = t;
60
61     end
62
63     %-----
64     % PDE DEFINITIONS
65     %  $c(x, t, u, ux) * u_t = x^{-m} * \frac{d}{dx}[x^m * f(x, t, u, ux)] + s$ 
66

```

```

67 % PDE coefficients for spin diffusion with microwave irradiation (DNP ON)
68 function [c,f,s] = pdex1pde(x,~,u,DuDx)
69     c = Cp * fc(x);
70     f = fD(x) * Cp * fc(x) * DuDx;
71     s = -(u - fe0(x)) * (1 / fT1(x)) * Cp * fc(x);
72 end
73
74 % PDE coefficients for spin diffusion without microwave irradiation (DNP OFF)
75 function [c,f,s] = pdex1pdenoDNP(x,~,u,DuDx)
76     c = Cp * fc(x);
77     f = fD(x) * Cp * fc(x) * DuDx;
78     s = -(u - depo(x)) * (1 / fT1(x)) * Cp * fc(x);
79 end
80
81 % Initial condition: zero polarization throughout the sample (DNP ON)
82 function u0 = pdex1ic(~)
83     u0 = 0;
84 end
85
86 % Initial condition: zero polarization throughout the sample (DNP OFF)
87 function u0 = pdex1icnoDNP(~)
88     u0 = 0;
89 end
90
91 % Zero-flux boundary conditions (DNP ON)
92 function [pl,ql,pr,qr] = pdex1bc(~,~,~,~,~)
93     pl = 0; ql = 1;
94     pr = 0; qr = 1;
95 end
96
97 % Zero-flux boundary conditions (DNP OFF)
98 function [pl,ql,pr,qr] = pdex1bcnoDNP(~,~,~,~,~)
99     pl = 0; ql = 1;
100    pr = 0; qr = 1;
101 end
102
103 %-----
104 % SPATIAL PROFILE FUNCTIONS (smooth tanh transitions)
105
106 % Longitudinal relaxation time (T1) profile
107 function T1 = fT1(y)
108     T1 = (T1in + T1out)/2 + (T1in - T1out)/2 * tanh(slope * (length_target - y));
109 end
110
111 % Steady-state polarization profile with microwave irradiation (DNP ON)
112 function epsilonzero = fe0(y)
113     epsilonzero = (epsilon0 * dep + 1)/2 + (epsilon0 * dep - 1)/2 * tanh(-slope *
114         ↪ (length_target - y));
115 end
116
117 % Observation (quenching) profile
118 function theta3 = theta0(y)
119     theta3 = (1 + theta)/2 + (1 - theta)/2 * tanh(slope * (length_target - y));
120 end
121
122 % Spin diffusion coefficient profile
123 function D = fD(y)
124     D = (Din + Dout)/2 + (Din - Dout)/2 * tanh(slope * (length_target - y));
125 end
126
127 % Equilibrium polarization profile without microwave irradiation (DNP OFF)
128 function depol = depo(y)
129     depol = (1 + dep)/2 + (1 - dep)/2 * tanh(slope * (length_target - y));
130 end
131
132 % Spin concentration profile
133 function c = fc(y)
134     c = (concX + concjuice)/2 + (concX - concjuice)/2 * tanh(slope * (length_target - y));
135 end

```

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